

Mishti Broor

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EDUCATION

Northwestern University

M.S. in Biomedical Engineering

Evanston, IL

September 2026 – June 2028

University of Illinois at Chicago (UIC)

Chicago, IL

B.S. in Biomedical Engineering – Concentration: Biomedical Imaging

January 2023 – May 2026

- Cumulative GPA: 3.80/4.00 | Magna Cum Laude

PROFESSIONAL EXPERIENCE

Research/Scientific Aide – Liu-CoMotion Lab (Dr. Chang Liu)

March 2025 – Present

University of Illinois at Chicago

Chicago, IL

- Engineered and fabricated novel EEG electrode attachments for unbraided and cornrowed hair using advanced CAD and resin 3D printing, ensuring compatibility with 10-20 and 10-05 international placement systems.
- Conducted a 30-subject, two-phase study utilizing BrainVision high-density EEG systems, IMUs, and force-sensor insoles for synchronized neural and biomechanical data collection across resting, overground, and treadmill-controlled walking trials.
- Lead experimental preparation and protocol execution, including digital electrode mapping, precision gel application, and impedance optimization below 15 kΩ to evaluate scalp proximity and participant comfort.
- Built, debugged, and validated a MATLAB–Python pipeline (individual alpha frequency estimation, FOOOF-corrected SNR modeling) benchmarking novel electrode performance against a conventional EEG cap across 17 participants, resolving critical data-integrity bugs to reach byte-identical reproducibility across runs.
- Co-authoring a peer-reviewed manuscript on the novel vs. conventional electrode comparison; reconcile participant data across processing stages, prepare statistical reporting and figures, and maintain IRB-compliant data management.

Clinical Immersion Intern (NIH-funded R25EB018239)

June 2025 – August 2025

University of Illinois at Chicago

Chicago, IL

- Completed a six-week ophthalmology immersion, shadowing clinical teams to identify workflow inefficiencies and unmet technology needs.
- Conducted patient-device interaction observations and collaborated with clinicians and engineers to create a comprehensive needs portfolio guiding senior capstone and medical device projects.
- Delivered a final report proposing innovation opportunities aligned with FDA regulatory and market considerations.

Technical Writer – Neurofounders

January 2026 – Present

- Produce in-depth analyses of wearable health technology, focusing on market trends and technical specifications of BCI products.

PROJECTS

High-Fidelity In-Vitro Spinal Cord Model

August 2025 – May 2026

- Award: 1st Place in BME; 4th Place Overall out of 200+ Projects, Engineering Expo (2026).
- Engineered an anatomically detailed 3D spinal cord model with high optical clarity, integrating MSLA-printed resin parts and custom silicone components for accurate Cerebrospinal Fluid (CSF) flow simulation.
- Utilized advanced Fusion 360 CAD modeling and mesh manipulation (Blender/MeshMixer) to optimize tissue architecture and physiological accuracy for fluid dynamics.

EEG Motor Imagery BCI Classifier

June 2026 – August 2026

- Built an EEG motor imagery decoding pipeline (MNE-Python, scikit-learn) on the PhysioNet EEG Motor Movement/Imagery Dataset (30 subjects, 64-channel EEG); Common Spatial Patterns (CSP) spatial filtering lifted single-subject accuracy from chance (~0.5) to up to 0.96 cross-validated accuracy.
- Ran per-subject statistical testing rather than relying on group averages, finding 5 of 30 subjects (17%) showed statistically significant above-chance decoding ($p < 0.05$), consistent with documented BCI illiteracy rates in the literature.
- Validated that decoding reflects genuine motor cortex activity via a channel-restriction control experiment, and diagnosed why per-subject CSP filters collapse to chance when pooled across subjects.

CONFERENCES

- 8th Annual BME Symposium (UIC), UIC Research Forum, & Chicago Area Undergraduate Research Symposium (CAURS)

April 2026

- Society for Neuroscience (SfN) Annual Meeting, Washington D.C.

November 2026

Poster Presentation: Design and Test Novel EEG Electrodes for Afro-Textured Hair.

CERTIFICATIONS

- MATLAB Signal Processing Onramp; Supervised Learning with scikit-learn; Statistical Thinking in Python; Python for MATLAB Users (DataCamp); Git & GitHub, Introduction through Intermediate (DataCamp); AI-Assisted Coding for Developers & Claude 101 (DataCamp)

PROFESSIONAL SKILLS

- Hardware & CAD:** CAD Modeling (SolidWorks, Fusion 360, Blender), 3D Printing (SLA), MeshMixer, Tinkercad, Arduino IDE
- Programming & AI:** Python, MATLAB (Signal Processing, Digital Filtering), Machine Learning (scikit-learn, MNE-Python), EEG Signal Decoding, Git/GitHub, Cursor, AI & Prompt Engineering
- Clinical & Methods:** EEG Data Acquisition & Analysis (BrainVision, 10-20/10-05), IMUs, IRB Guidelines, Workflow Optimization, Technical Writing